

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

B.TECH (IIT, 4th Sem), END-SEMESTER EXAM (08-May-2025)

SUBJECT: MICROPROCESSORS & COMPUTER ARCHITECTURE

COURSE CODE: IIPC 209

Maximum Marks: 50

Time: 3 Hrs

Attempt one question out of Q.5 and Q.6

Assume necessary info. if not given

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1. (a) Discuss need of address decoding, and different address decoding techniques. Find memory map of 20KB memory chip. [4]
(b) Draw circuit to interface two common anode seven segment display (SSD) with F2 and F3 addresses. Write instruction to write "97" at the SSD LED port. Use 74LS373, 74LS138. Draw timing diagram for the OUT instruction used in the above program. [6]
 2. (a) Explain the instructions LXI, DAA, LHL, STAX, RAR with proper examples. [5]
(b) In a home guard system, a door sensor is connected to the PORT A, window sensor is connected to the PORT B, and the siren is connected to PORT C. Write 8085 assembly program to monitor the door and window sensor. When any one gets open, sound the siren by sending a square wave of 100ms time period. (Port A address is 5670H). Draw complete circuit diagram. [5]
 3. (a) Interface 4k bytes RAM and 8k bytes ROM to 8051 microcontroller in such a way that starting address of RAM is 1000H and ROM is C000H. [4]
(b) Store the ASCII values of the first 10 capital letters of the alphabet in ROM locations. Bring these 10 values to RAM locations starting from 40H. [3]
(c) Explain 8051 memory organisation. [3]
 4. (a) We have 2 designs D1 and D2 for a synchronous pipeline processor. D1 has 5 stage pipeline with execution time of 3 ns, 2 ns, 4 ns, 2 ns and 3 ns. While the design D2 has 8 pipeline stages each with 2 ns execution time. How much time can be saved using design D2 over design D1 for executing 100 instructions? [2]
(b) Explain Data hazards and ways to overcome using proper diagram. [4]

(c) Discuss Hardwired control, Micro programmed control circuit in details. Suggest suitable processor architecture for the both. [4]

5. (a) Discuss data transfer using strobe and handshaking mode using proper timing diagram. [4]

(b) How processor manages Interrupt priority for serially connected and parallel connected devices that request an interrupt? [3]

(c) Draw the block diagram for the hardware that implements the following statement $x + yz: AR \leftarrow AR + BR$ Where AR and BR are two n-bit registers and x, y, and z are control variables. Include the logic gates for the control function [3]

OR

6. (a) Write Control sequencing for ADD R1, [R3] and discuss control path of the same instruction considering single bus organisation. [4]

(b) Differentiate between RISC and CISC architectures with advantages and disadvantages. [2]

(c) Short note on [4]

I. One and three address instructions with examples.

II. Asynchronous Serial data transfer protocol.
